Exp. No: 5 Electric Potential Difference between Two Points in Free Space

Aim:

To write program for Electric potential difference between two points in free space medium using MATLAB.

Objectives:

- To determine the electric potential difference between two points separated by a distance R in free space.
- To plot the variation of potential difference with separation distance.
- To relate the potential gradient to electric field intensity for validation.

Test case (i) Determination of electric potential difference between two points for a given charge

```
% Clear workspace and command window

clc;

clear;

close all;


%% Constants & User Inputs

epsilon_0 = 8.854e-12;              % Permittivity of free space (F/m)

k = 1 / (4 * pi * epsilon_0);       % Coulomb constant (N·m^2/C^2)


q = input('Enter the q value: ');

ax = input('Enter the value of ax=');

ay = input('Enter the value of ay=');


% Distance and potential at point 1

r1 = hypot(ax, ay);                 % r1 = sqrt(ax^2 + ay^2)

v1 = (k * q) / r1;

disp('The value of v1');

disp(v1);
```

```
bx = input('Enter the value of bx=');
by = input('Enter the value of by=');

% Distance and potential at point 2
r2 = hypot(bx, by);                 % r2 = sqrt(bx^2 + by^2)
v2 = (k * q) / r2;
disp('The value of v2');
disp(v2);

% Potential difference between point 1 and point 2
v_diff = v1 - v2;
disp('The electric potential difference is');
disp(v_diff);

%% Plotting
% 1. Potential V versus radial distance r
r_grid = linspace(0.1, 10, 50);
V_grid = (k * q) ./ r_grid;

figure('Name', 'Radial Potential');
plot(r_grid, V_grid, 'LineWidth', 2);
hold on;
plot(r1, v1, 'ro', 'MarkerFaceColor', 'r');
grid on;
xlabel('Radial distance r (m)');
ylabel('Potential V (V)');
title('Potential V = kq/r');
legend('V(r)', 'V1 at r1');
```

```
% 2. Potential difference V1 - V(r) versus radial distance r
figure('Name', 'Potential Difference');
delta_V = k * q * (1/r1 - 1 ./ r_grid);
plot(r_grid, delta_V, 'LineWidth', 2);
grid on;
xlabel('Radial distance r (m)');
ylabel('Potential difference V1 - V(r) (V)');
title('Potential difference between r1 and r');

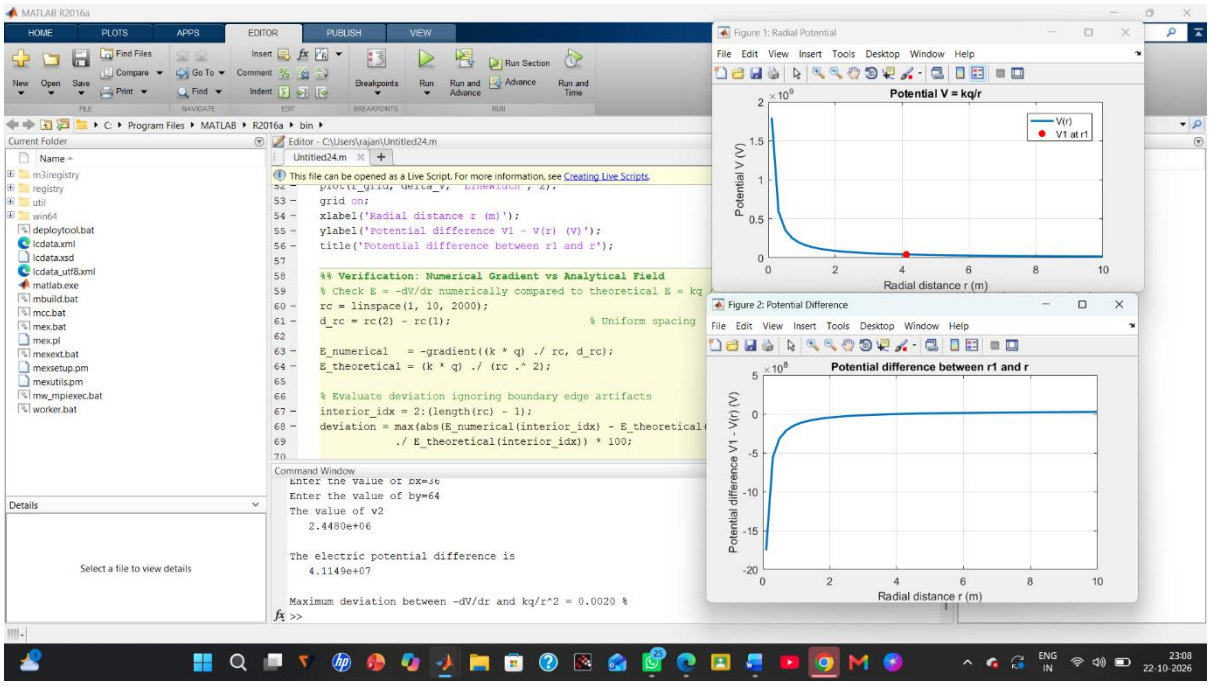
%% Verification: Numerical Gradient vs Analytical Field
% Check E = -dV/dr numerically compared to theoretical E = kq / r^2
rc = linspace(1, 10, 2000);
d_rc = rc(2) - rc(1);                % Uniform spacing

E_numerical  = -gradient((k * q) ./ rc, d_rc);
E_theoretical = (k * q) ./ (rc .^ 2);

% Evaluate deviation ignoring boundary edge artifacts
interior_idx = 2:(length(rc) - 1);
deviation = max(abs(E_numerical(interior_idx) - E_theoretical(interior_idx)) ...
        ./ E_theoretical(interior_idx)) * 100;

fprintf('Maximum deviation between -dV/dr and kq/r^2 = %.4f %%\n', deviation);
```

Exp. No: 5 Electric Potential Difference between Two Points in Free Space



Test case (ii) A positive charge with fixed r_1 = 1 m and r_2 varying.

```
clc;

epsilon=8.854e-12;

k=1/(4*pi*epsilon);

q=1e-6;

r1=1;

r2=linspace(1.1,10,100);

V=k*q*(1/r1 - 1./r2);

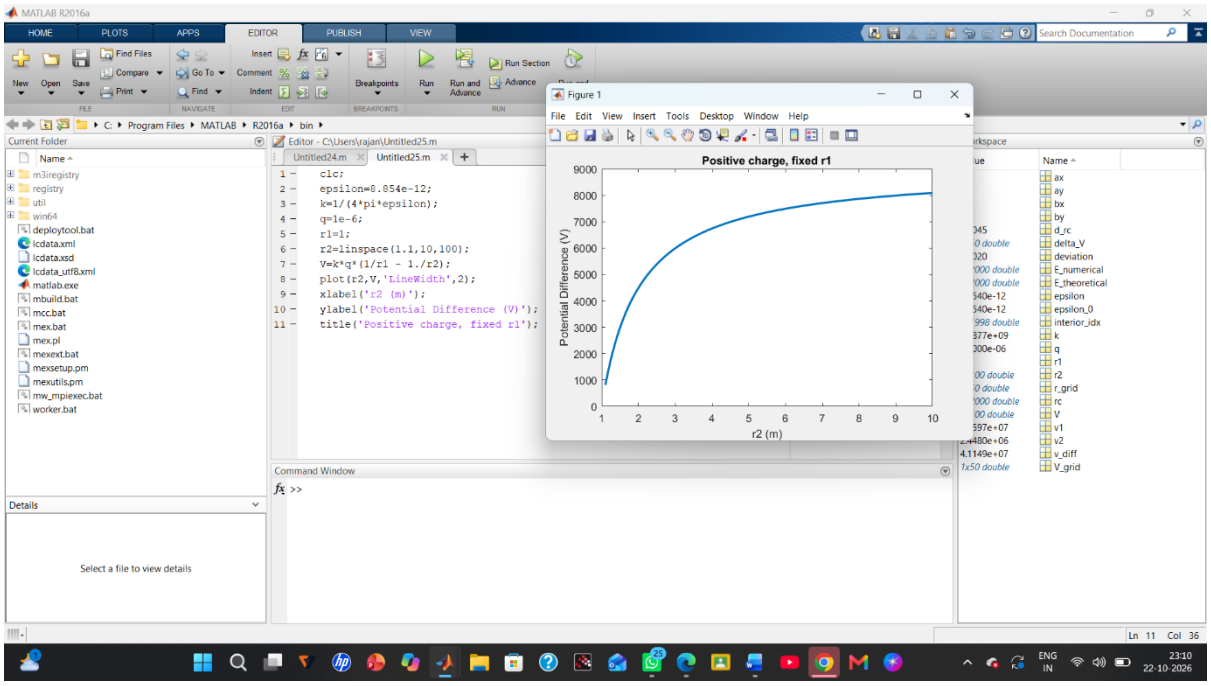
plot(r2,V,'LineWidth',2);

xlabel('r2 (m)');

ylabel('Potential Difference (V)');

title('Positive charge, fixed r1');
```

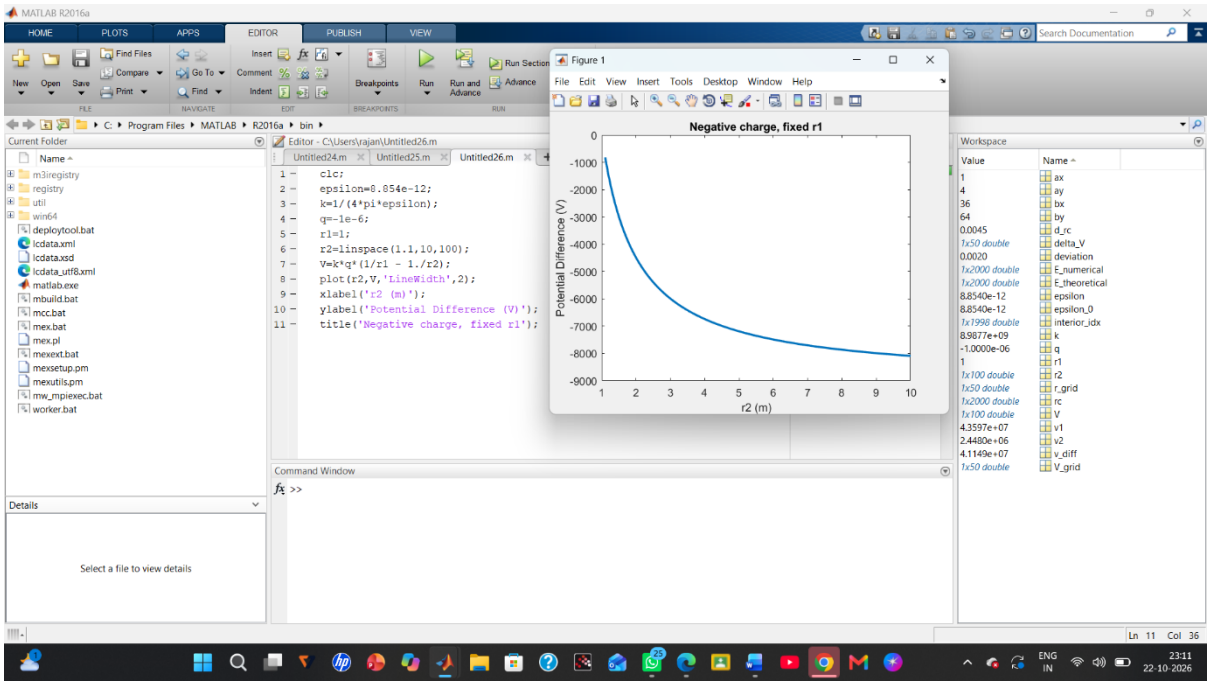
Exp. No: 5 Electric Potential Difference between Two Points in Free Space



Test case (iii) A negative charge with fixed r_1 and varying r_2:

```
clc;
epsilon=8.854e-12;
k=1/(4*pi*epsilon);
q=-1e-6;
r1=1;
r2=linspace(1.1,10,100);
V=k*q*(1/r1 - 1./r2);
plot(r2,V,'LineWidth',2);
xlabel('r2 (m)');
ylabel('Potential Difference (V)');
title('Negative charge, fixed r1');
```

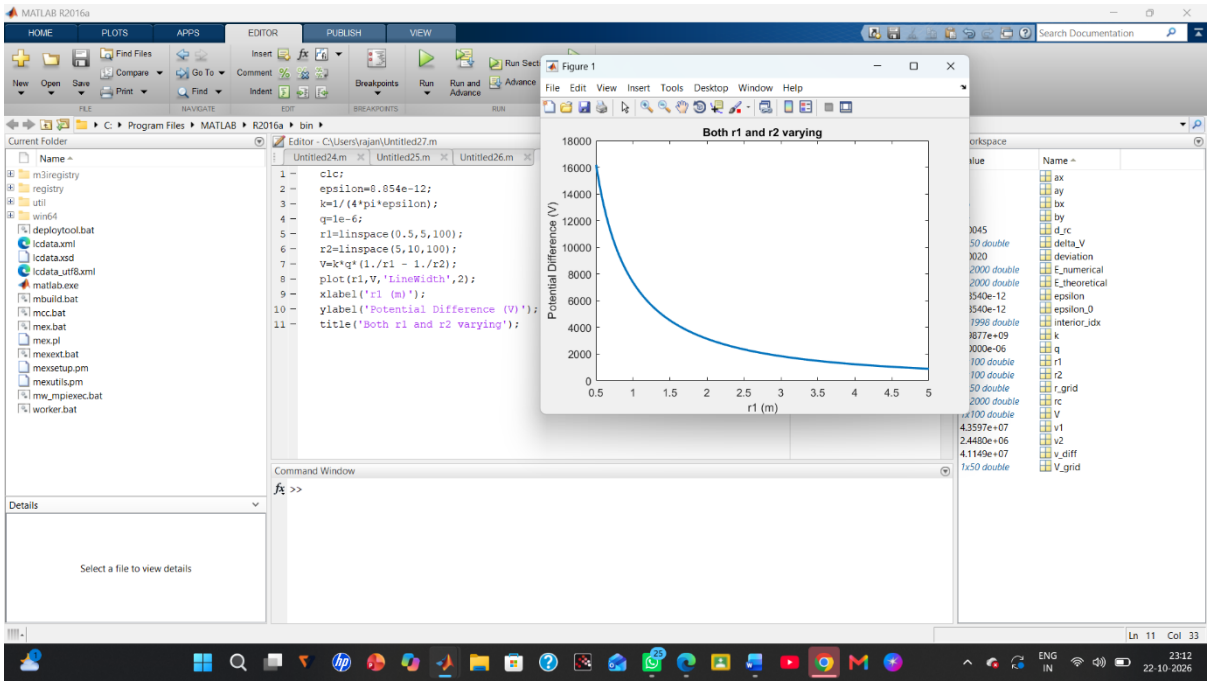
Exp. No: 5 Electric Potential Difference between Two Points in Free Space



Test case (iv) Both r_1 and r_2 varying with positive charge 3:

```
clc;
epsilon=8.854e-12;
k=1/(4*pi*epsilon);
q=1e-6;
r1=linspace(0.5,5,100);
r2=linspace(5,10,100);
V=k*q*(1./r1 - 1./r2);
plot(r1,V,'LineWidth',2);
xlabel('r1 (m)');
ylabel('Potential Difference (V)');
title('Both r1 and r2 varying');
```

Exp. No: 5 Electric Potential Difference between Two Points in Free Space



Exp. No: 5 Electric Potential Difference between Two Points in Free Space

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)
1	Define charge, reference and observatio n points	Define charge, reference and observation points	Correctly identifies charge distribution, point locations, distance and reference convention.	Setup is correct with minor clarifica tion needed.	Some input or reference details are unclear.	Uses an incorrect setup or reference.	25
2	Calculate potential and potential difference	Calculate potential and potential difference	Calculates potential at both points and signed difference correctly with units.	Results are mostly correct with minor arithmet ic errors.	Partial calculations require support.	Potential calculations are incorrect or absent.	25
3	Verify potential relationship	Verify potential relationship	Checks results against analytical equations and, where relevant, the relation between E and potential.	Validati on is mostly correct with minor omissio ns.	Validation is incomplete.	No valid theoretical check is provided.	30
4	Interpret and report findings	Interpret and report findings	Explains the sign and physical meaning of potential difference with clear tables/conclus ion	Explana tion is correct with minor omissio ns.	Explanation is limited or unclear.	Cannot explain or document the result.	20
Tota l							100